

**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED
BACKEND DEVELOPMENT (25CS1302E)**

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No- 8	Team No- 17	Project Title- School Bus Tracking & Student Safety Portal
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Abstract

The School Bus Tracking & Student Safety Portal is a web-based system designed to improve the safety, visibility, and management of school transportation. The system addresses the problem of limited real-time visibility of school buses and uncertainty regarding student boarding and drop-off activities.

The proposed solution provides parents with live school-bus tracking and timely notifications about their child's boarding and exit events. GPS integration enables real-time monitoring of bus locations, while the Bus In-charge Portal allows the in-charge to record student boarding and drop-off events accurately. Parents receive notifications through Firebase Cloud Messaging, allowing them to stay informed about their child's transportation status.

The system also includes route and stop management, driver management, trip history, and reporting features to support efficient administration of school transportation. A React.js-based frontend provides an interactive user interface, while Node.js and Express.js handle backend services and communication with the MySQL database. Google Maps is used for map-based tracking and route visualization.

Overall, the system aims to provide a centralized, reliable, and user-friendly platform that enhances student safety, parent awareness, and school bus management.

1. Team Details

Roll Number(s)	Name of the Student	Signature
2520030106	D. Mohit Venkat Sai	
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2. Frameworks / Platforms / Software Used

- **React.js** (*Frontend*)
- **Node.js** (*Backend*)
- **Express.js** (*Backend APIs*)
- **MySQL** (*Database*)

- **GPS Device API** (*Live Bus Tracking*)
- **Firebase Cloud Messaging (FCM)** (*Parent Notifications*)
- **Google Maps API** (*Bus Route & Location Visualization*)

Guide Name and Signature (remarks if any) _____